

外 文 翻 译

毕业设计题目：面向服饰纹理替换的智能重绘方法研究与系统开发

原文 1：Normal-guided Garment UV Prediction for Human Re-texturing

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Normal-guided Garment UV Prediction for Human Re-texturing

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Abstract: Clothes undergo complex geometric deformations, which lead to appearance changes. To edit human videos in a physically plausible way, a texture map must take into account not only the garment transformation induced by the body movements and clothes fitting, but also its 3D fine-grained surface geometry. This poses, however, a new challenge of 3D reconstruction of dynamic clothes from an image or a video. In this paper, we show that it is possible to edit dressed human images and videos without 3D reconstruction. We estimate a geometry aware texture map between the garment region in an image and the texture space, a.k.a, UV map. Our UV map is designed to preserve isometry with respect to the underlying 3D surface by making use of the 3D surface normals predicted from the image. Our approach captures the underlying geometry of the garment in a self-supervised way, requiring no ground truth annotation of UV maps and can be readily extended to predict temporally coherent UV maps. We demonstrate that our method outperforms the state-of-the-art human UV map estimation approaches on both real and synthetic data.

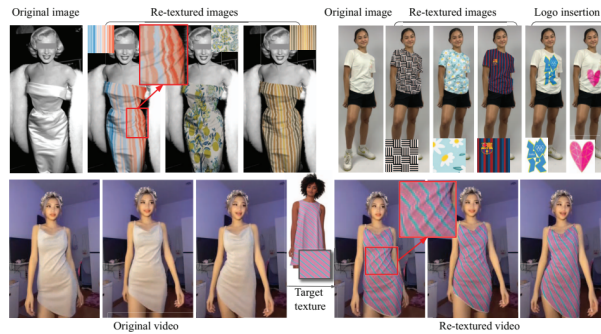


Figure 1. demonstrates geometry-aware texture editing for images and videos; this paper puts forward a 2 novel approach to predict a geometry-aware texture (UV) map of a garment from an image (first row) and a video (second row), where the predicted UV map preserves isometry between texture space and 3D surface by leveraging 3D surface normals, and further, the approach ensures temporal consistency in the predicted UV map across frames in a video, thus achieving physically plausible human appearance editing, with the project website available at yasamin.page/normal-guided-uv.

1. Introduction

While browsing online clothing shops, have you ever wondered how the appearance of a dress of interest would look on you as if you were in a fitting room given your dress with a similar shape? A key technology to enable generating such visual experiences is photorealistic re-texturing—editing the texture of clothes in response to the subject’s movement in the presented images or videos in a geometrically and temporally coherent way. Over the past few years, there has been a significant advancement in the image and video editing technologies, such as inserting advertising logos on videos of moving cars or applying face makeup on social media. However, such editing approaches designed for rigid or semi-rigid surfaces are not suitable for garments that undergo complex secondary motion with respect to the underlying body. For example, the fine wrinkles of the dress in Figure 29 result in complex warps in texture over time. In this paper, we present a new method to edit the appearance of a garment in a given image or video by taking into account its fine-grained geometric deformation.

Previous works address photorealistic texture editing in two ways. (1) **3D reconstruction and rendering:** these approaches can achieve high-fidelity texture editing given highly accurate 3D geometry. On the other side of the coin, their performance is dictated by the quality of the 3D reconstruction. While the 3D geometry of the garment can be learned from paired human appearance data, e.g., human modeling repositories with 3D meshes and renderings, due to the scarcity of such data, it often cannot generalize well on unseen real images and videos. (2) **Direct texture mapping:** by estimating dense UV map, these methods can bypass the procedure of 3D reconstruction. However, they usually lack of geometry details and only capture the underlying human body, thus, not applicable for editing garments. Moreover, when applied to videos, visual artifacts of editing become more salient since they are not aware of underlying deformation of the garment’s 3D geometry.

We design our method to enjoy the advantages of both two approaches: preserving realistic details in UV mapping while circumventing 3D reconstruction. Our key insight is that the fundamental geometric property of isometry can be imposed into UV map estimation via the 3D surface normals predicted from an image. We formulate a geometric relationship between the UV map and surface normals in the form of a set of partial differential equations.

Our method takes as input an image or video, its surface normal prediction, and dense optical flow (for video), and outputs the geometry aware UV map estimate. The UV map is modeled by a multi-layer perceptron that can predict UV coordinates given a pixel location in

an image. We note that the UV map is defined up to the choice of a reference coordinate frame. To disambiguate this, we condition the neural network with a pre-defined proxy UV map (e.g., DensePose). We use the isometry constraints as a loss to optimize the UV map. Further, for a video, we leverage the per-frame image feature to correlate the UV coordinates of the pixels across time using optical flow.

Our contributions can be concluded in three aspects:

- A novel formulation that captures the geometric relationship between the 3D surface normals and the UV map by the isometry constraint, which eliminates the requirement of 3D reconstruction and ground truth UV map;
- A neural network design that learns to predict temporally coherent UV map for the frames by correlating per-frame image features;
- Stronger performance compared to existing re-texturing methods and compelling results on a wide range of real-world imagery, as demonstrated in Table 4.

2. Related Work

Our work lies at the intersection of human UV map prediction from images and neural UV map optimization.

2.1. Human Dense UV Map Estimation

A seminal work of DensePose learns to predict a UV map of humans presented in an image, which opens a new opportunity to edit the appearance of a person without 3D reconstruction. A series of subsequent works bring out a number of applications for human tracking. However, due to their representation specific to the body surface, they exhibit fundamental limitations in expressing highly deformable loose clothing such as skirts and dresses. To address this challenge, recent approaches leverage multitask learning or incorporate geodesic distance to learn UV maps. BodyMap incorporates the Vision Transformers to learn per-pixel image features on a continuous body surface that handles loose clothes, different hairstyles, and occlusion. TemporalUV focuses on handling garments by extrapolating the initial DensePose estimates and leveraging image features obtained from an input video to obtain a UV aligned with the garment boundary. Despite their promise, the visual artifacts persist due to a lack of understanding the underlying 3D geometry. Unlike previous approaches, we design our framework such that the

resulting UV map satisfies the fundamental geometric property of isometry, which results in physically plausible re-texturing.

2.2. Neural UV Optimization from Videos

Another line of work resorts to a layered UV map, capturing the geometry to some degree by incorporating video decomposition to optimize the UV coordinates of the foreground and background based on the observed motion. Kasten et al. unwrap a video into a set of layered 2D atlases where for each pixel in the video, its corresponding 2D coordinate in each of the atlases is predicted. Ye et al. proposes a global sprite image that can group the distinct motion trajectories because the collective object structure has a consistent appearance throughout time. While preserving the temporal coherency and maintaining some coarse UV deformations related to arm or leg movements during a sequence, these methods fall short of capturing micro deformations like wrinkles in the clothing. Furthermore, these techniques cannot represent UV mapping for an image and can only be applied to videos. Unlike these methods, our surface normal conditioned UV map is highly sensitive to small geometric details and 3D surface deformations, which can be optimized not only for a video but also for single images.

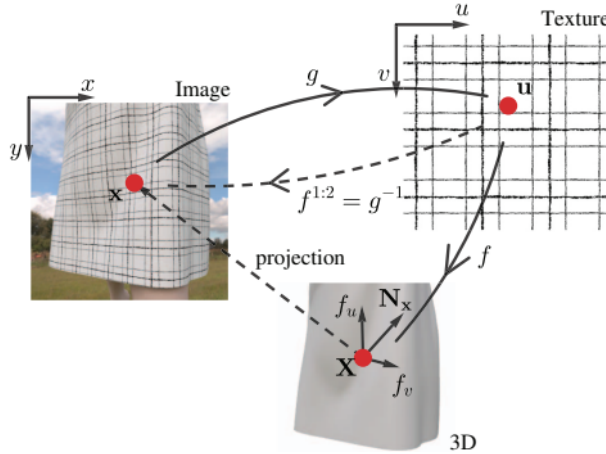


Figure 2. Illustration of the mapping $g(x) = u$ that maps a pixel location in the image space to a point in the UV space of the garment.

3. Method

Our goal is to obtain a continuous texture mapping, which allows editing the appearance of dynamic garments. We leverage the geometric property of isometry to constrain the UV map in the form of partial differential equations. We solve these partial differential equations by

optimizing a neural network to generate a geometry aware UV map.

3.1. Texture Mapping without 3D Reconstruction

Consider a mapping $g(x) = u$ that maps a pixel location $x = (x, y) \in \mathbb{R}^2$ in the image space that belongs to a garment of interest to a point $u = (u, v) \in \mathbb{R}^2$ in the UV space of the garment. The goal of our work is to find such a mapping g that takes into account the local surface geometry measured by the surface normal predicted at x . We denote the predicted 3D surface normal of x in the camera space as $N_x \in \mathbb{S}^2$.

Let us define an isometric map from the UV texture map to the 3D surface, $f(u) = X$. This is the fundamental property of a non-stretchable cloth texture mapping:

$$\|f_u\| = \|f_v\| = 1, \quad f_u^T f_v = 0, \quad (1)$$

where f_u and f_v are the partial derivatives of f with respect to u and v , respectively. Geometrically, f_u and f_v are the tangential vectors on the 3D surface where their cross product forms the surface normal:

$$\tilde{N}_x = f_u \times f_v = f_u(g(x)) \times f_v(g(x)) \quad (2)$$

where $\tilde{N}_x \in \mathbb{S}^2$ is the surface normal at X corresponding to x . We can find the UV mapping g by matching the surface normal $\tilde{N}_x$ derived by Equation (2) and the surface normal predicted from the image N_x :

$$\begin{aligned} \text{Minimize}(\theta_g, \theta_f) \quad & \sum_x \|f_u(g(x)) \times f_v(g(x)) - N_x\|^2, \\ \text{s.t.} \quad & \|f_u\| = \|f_v\| = 1, f_u^T f_v = 0, \end{aligned} \quad (3)$$

where θ_g and θ_f are the parameters of the function g and f , respectively. To solve this without full 3D reconstruction, we assume orthographic projection, i.e., $(x, y) = (X, Y)$. This allows us to express the 3D derivatives using the pixel coordinates:

$$g = (f^{1:2})^{-1}, \quad (4)$$

where $f^{1:2}$ is the first two elements of f . Using the inverse function theorem, the Jacobian matrix J_g is related to f via:

$$f^{1:2} \circ g(x) = x \rightarrow J_g = (J_{f^{1:2}})^{-1}, \quad (5)$$

By eliminating f , Equation (2) can be re-written as the following constraints:

$$\|g_x\| = \sqrt{1 + \frac{\tilde{n}_x^2}{\tilde{n}_z^2}}, \quad \|g_y\| = \sqrt{1 + \frac{\tilde{n}_y^2}{\tilde{n}_z^2}}, \quad g_x^T g_y = \frac{\tilde{n}_x \tilde{n}_y}{\tilde{n}_z^2}, \quad (6)$$

where $\tilde{N}_x = [\tilde{n}_x \ \tilde{n}_y \ \tilde{n}_z]^T$. As illustrated in Figure 31, this captures the isometric relationship between the surface and the UV map. This leads to the geometric loss function:

$$\begin{aligned} \mathcal{L}_{\text{geo}}(\theta_g) = \sum_x & \left(\left(\frac{\partial u}{\partial x} \Big|_x \right)^2 + \left(\frac{\partial v}{\partial x} \Big|_x \right)^2 - 1 - \frac{n_x^2}{n_z^2} \right)^2 \\ & + \left(\left(\frac{\partial u}{\partial y} \Big|_x \right)^2 + \left(\frac{\partial v}{\partial y} \Big|_x \right)^2 - 1 - \frac{n_y^2}{n_z^2} \right)^2 \\ & + \left(\frac{\partial u}{\partial x} \Big|_x \frac{\partial u}{\partial y} \Big|_x + \frac{\partial v}{\partial x} \Big|_x \frac{\partial v}{\partial y} \Big|_x - \frac{n_x n_y}{n_z^2} \right)^2, \end{aligned} \quad (7)$$

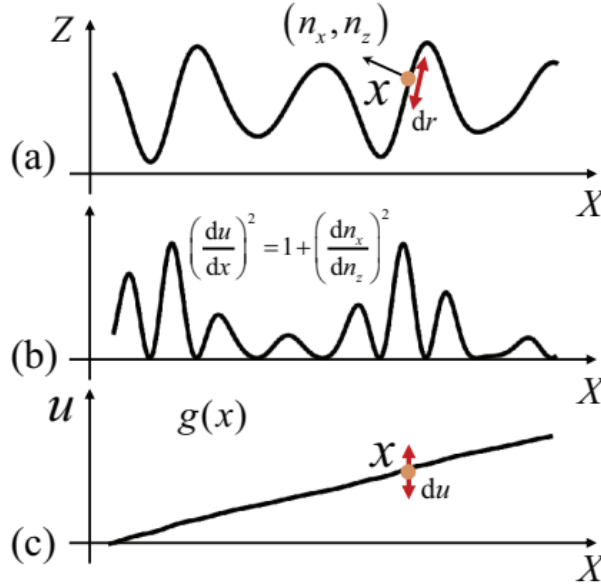


Figure 3. 2D simplification of UV map estimation. We illustrate the isometric relationship between (a) the surface and (c) the UV map. The length of the curve in XZ axis needs to be preserved when mapping to $g(x)$, i.e., $|du| = |dr| = \sqrt{dx^2 + dz^2}$ (arc length preservation). This relationship can be re-written as (b) a partial differential equation using Equation (6) in terms of its surface normal and the inverse of the spatial derivative of g .

3.2. Self-supervised Learning of Texture Mapping

To resolve the ambiguity of the reference coordinate frame, we find g such that $g \approx g'$ where g' is a pre-defined proxy map:

$$\mathcal{L}_{\text{prox}}(\theta_g) = \sum_x \|g'(x) - g(x)\|^2 \quad (8)$$

In practice, we use an extended DensePose as the proxy. Further, to ensure the surface normals point toward the camera (+Z direction), we add:

$$\mathcal{L}_z(\theta_g) = \sum_x \max(0, -\det(J_g|_x)) \quad (9)$$

For video, we ensure temporal consistency using optical flow:

$$\mathcal{L}_{\text{tmp}}(\theta_g) = \sum_{i,j} \sum_{x_i} \|g(x_i, f_{x_i}) - g(x_j, f_{x_j})\|^2 \quad (10)$$

The overall loss function is:

$$\mathcal{L}(\theta_g) = \mathcal{L}_{\text{geo}} + \lambda_{\text{prox}} \mathcal{L}_{\text{prox}} + \lambda_z \mathcal{L}_z + \lambda_{\text{tmp}} \mathcal{L}_{\text{tmp}} \quad (11)$$

where $\lambda_{\text{tmp}} = 0$ for single images.

3.3. Implementation Details

We model g using a 12-layer multi-layer perceptron (as shown in Figure 32) with ReLU activations. Pixel coordinates are Fourier-based positionally encoded $\gamma : \mathbb{R}^2 \rightarrow \mathbb{R}^{128}$. For videos (Figure 33), we use ResNet to extract 256-dimensional image features f_x . We use the Adam optimizer with a batch size of 2048 and learning rate of 10^{-4} . Our method takes 20 minutes for a video of 82 frames, significantly faster than existing baselines compared in Table 4.

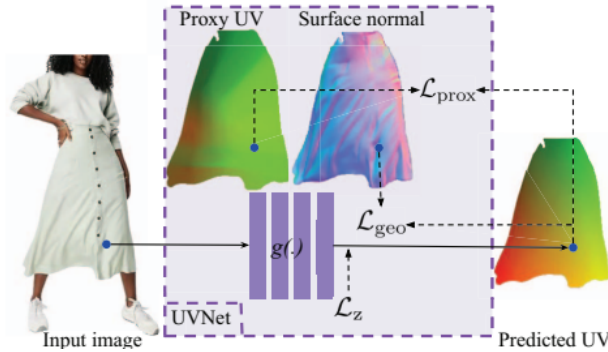


Figure 4. Single image framework. For each pixel location on a garment x , we predict the UV coordinate u using a multi-layer perceptron. We enforce isometry to the UV map by matching with the predicted 3D surface normals. $\mathcal{L}_z$ ensures that the predicted surface normals point toward the camera. To disambiguate the frame of reference of the UV map, we use pre-defined proxy map (e.g., DensePose extrapolation) by applying $\mathcal{L}_{\text{prox}}$.

4. Evaluation

We evaluate our method both quantitatively and qualitatively on real images as well as synthetic data with ground truth UV map. We also compare with the state-of-the-art methods on human UV map estimation and video UV map optimization approaches.

4.1. Evaluation Datasets and Metrics

Evaluation Datasets: We evaluate our method using: (1) five synthetic video sequences of simulated dress and T-shirt garments with random texture patterns over 700 frames; (2) ten real videos from Fashion Video dataset; (3) TikTok dataset and various YouTube videos as well as in-the-wild internet images.

Evaluation Metrics: We use five metrics: (1) **UV error:** For synthetic data, we report absolute UV error in texture space. We use Procrustes analysis to align the texture map. Mean squared error is reported in metric scale (Table 4). (2) **Average Precision percentage (AP):** Computed on all pixels; a prediction is correct if the UV error is lower than a threshold. Thresholds from 1 to 15 cm are visualized in Figure 18. (3) **Photometric error:** We warp the first frame of a video to the rest using UV estimates and report RGB difference (Table 4). (4) **Geometric error:** We report $\mathcal{L}_{\text{geo}}$ to show how the UV map follows surface normal estimates. (5) **Temporal error:** We evaluate temporal coherency by reporting the $\mathcal{L}_{\text{tmp}}$ error.

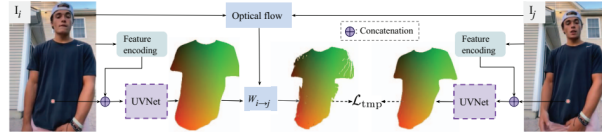


Figure 5. Video framework. We apply temporal coherence in the predicted UV maps using optical flow. With the optical flow $W_{i \rightarrow j}$, we match the UV prediction of i^{th} frame with j^{th} via $\mathcal{L}_{tmp}$.



Figure 6. The impact of $\mathcal{L}_{prox}$ in UV prediction and retexturing.

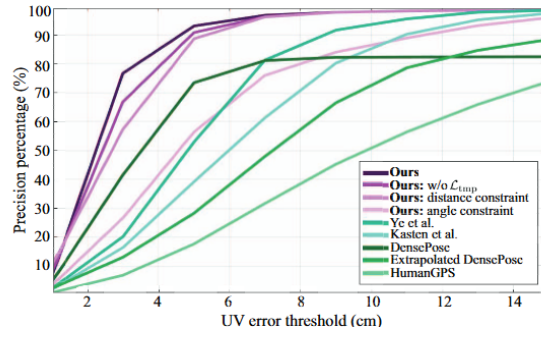


Figure 7. Average precision. We compute the average of pixels with a UV error higher than a threshold (1-15cm) on our method, the baseline methods, and our ablation experiments on ground truth dress sequences.

Table 1. Quantitative Results. UV.error (cm), photo. error (RGB difference), geo. error ($\mathcal{L}_{geo}$), and tmp. error ($\mathcal{L}_{tmp}$) (image space pixel distance) respectively (mean $\pm$ std).

Method	GT dress sequences [45]		GT T-shirt sequences [45]		Real Fashion sequences [59]	
	UV. error (cm)	photo. error	UV. error (cm)	photo. error	geo. error ($\mathcal{L}_{geo}$)	tmp. error
DensePose [18]	17.27 $\pm$ 3.76	52.43 $\pm$ 10.20	7.48 $\pm$ 0.52	34.19 $\pm$ 7.01	1.26 $\pm$ 0.47	8.12 $\pm$ 2.70
Extrapolated DensePose [18, 50]	8.61 $\pm$ 0.76	18.29 $\pm$ 3.24	5.34 $\pm$ 0.63	18.86 $\pm$ 4.55	0.52 $\pm$ 0.06	4.62 $\pm$ 0.73
HumanGPS [47]	11.97 $\pm$ 2.06	97.41 $\pm$ 30.87	7.53 $\pm$ 0.43	108.10 $\pm$ 34.89	1.27 $\pm$ 0.69	51.40 $\pm$ 41.66
Kasten et al. [25]	7.06 $\pm$ 0.60	13.07 $\pm$ 2.79	6.64 $\pm$ 0.63	14.54 $\pm$ 4.26	0.56 $\pm$ 0.10	2.30 $\pm$ 0.70
Ye et al. [57]	5.56 $\pm$ 0.29	33.57 $\pm$ 10.27	5.75 $\pm$ 0.20	19.22 $\pm$ 5.14	0.71 $\pm$ 0.03	1.65 $\pm$ 0.15
Ours	3.16 $\pm$ 0.28	7.54 $\pm$ 2.04	3.58 $\pm$ 0.27	11.28 $\pm$ 2.01	0.07 $\pm$ 0.03	1.50 $\pm$ 0.23

Table 2. Ablation study on dress sequences. UV. error (cm), photo. error (RGB difference) (mean $\pm$ std).

Method	UV. error (cm)	photo. error (RGB)
Ours	3.16 $\pm$ 0.28	7.54 $\pm$ 2.04
Distance constraint	3.44 $\pm$ 0.25	8.29 $\pm$ 1.99
Angle constraint	5.96 $\pm$ 0.33	7.68 $\pm$ 1.37
W/o $\mathcal{L}_{tmp}$	3.22 $\pm$ 0.24	14.67 $\pm$ 4.82
W/o $\mathcal{L}_{prox}$	3.24 $\pm$ 0.35	8.46 $\pm$ 2.19

4.2. Ablation Study

We conduct an ablation study to analyze the impact of the distance and angle constraints, as summarized in Table 5 and Figure 18. Our final method achieves the best performance in UV and photometric error. As expected, removing the temporal consistency term ($\mathcal{L}_{tmp}$) results in poor photometric error since consistency across frames is not enforced. The role of $\mathcal{L}_{prox}$ is to disambiguate equivalent UV maps; as shown in Figure 34, re-texturing without $\mathcal{L}_{prox}$ can result in arbitrary orientation across subjects.

4.3. Qualitative Results and User Study

Table 3. User study.

Method	User score
DensePose	4.09 ± 1.94
Proxy DensePose	3.23 ± 1.86
HumanGPS	2.28 ± 2.12
Kasten et al.	5.52 ± 1.88
Ye et al.	7.66 ± 1.27
Ours	9.57 ± 0.81

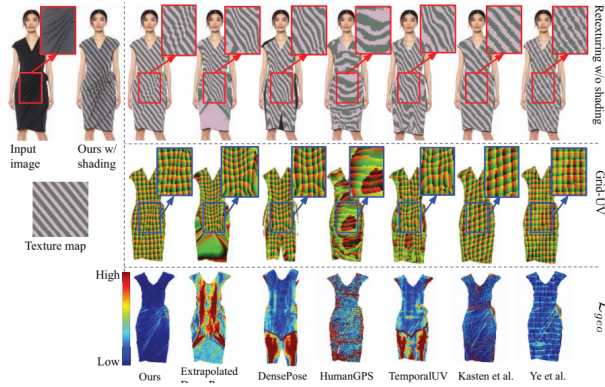


Figure 8. Qualitative comparison. We compare ours with our initial UV (extrapolated DensePose), DensePose, HumanGPS, Kastenn et al., and Ye et al. in image re-texturing and UV map grid visualization.



Figure 9. Limitation. Due to lack of 3D reconstructed surface, our method cannot handle folds that introduce texture discontinuity. Our texture map is continuous around the folds, which is physically incorrect.

To show our results qualitatively, we visualize re-texturing examples and grid UV illustrations to depict performance in preserving high-frequency details. Figure 19 illustrates results on the Fashion video sequence, while Figure 22 shows performance on general videos and images. Our method captures fine-grained surface details and remains temporally coherent.

We also conducted a user study ($n=21$) to rate realism (1: unrealistic to 10: most realistic). As summarized in Table 6, our method receives the highest score from users (9.57 ± 0.81) compared to other baselines.

5. Discussion

This paper presents a novel approach to predict high-quality UV maps by preserving geometric details from images and videos. We leverage isometry encoded in 3D surface normals to optimize the UV map via partial differential equations. By integrating optical flow, we generalize the method to videos, resulting in temporally coherent editing.



Figure 10. Limitation. Limitations on textured garment.



Figure 11. Results on images and videos. Our method can produce compelling texture editing given images and videos that preserve geometric details.

Limitations As discussed in Section 3.1, our method assumes a one-to-one correspondence between 3D surface geometry and the image. However, this assumption does not hold when there is a fold where a region of the 3D surface is not visible. Figure 20 illustrates this limitation: where there are folds in a skirt, the texture should be discontinuous, but our method produces a continuous texture rendering due to the lack of 3D reconstructed geometry, which is physically incorrect.

Another limitation concerns garments with highly contrasted textures. The shading operation is often biased toward color contrast, resulting in an unrealistic appearance near textured regions. Figure 21 demonstrates that despite reasonable UV prediction, the final appearance near complex patterns remains a challenge. Improving the shading operation is beyond the scope of this work and remains as future work.

译文 1

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基于法向引导的服装 UV 预测用于人体重贴图

摘要：服装经历复杂的几何形变，进而导致外观变化。要以物理上合理的方式编辑人体视频，纹理贴图不仅需要考虑由身体运动和服装贴合引起的衣物形变，还需考虑其三维细粒度表面几何形态。然而，这提出了从图像或视频中进行动态服装三维重建的新挑战。本文证明，无需三维重建即可实现着装人体图像与视频的编辑。我们通过建立图像中服装区域与纹理空间（即 UV 贴图）之间的几何感知纹理映射来实现这一目标。该 UV 贴图利用从图像预测的三维表面法线，旨在保持与底层三维表面的等距特性。我们的方法以自监督方式捕捉服装的基础几何结构，无需 UV 贴图的真实标注数据，并可轻松扩展至预测时序连贯的 UV 贴图。实验证明，在真实与合成数据上，我们的方法均优于当前最先进的人体 UV 贴图估计方案。

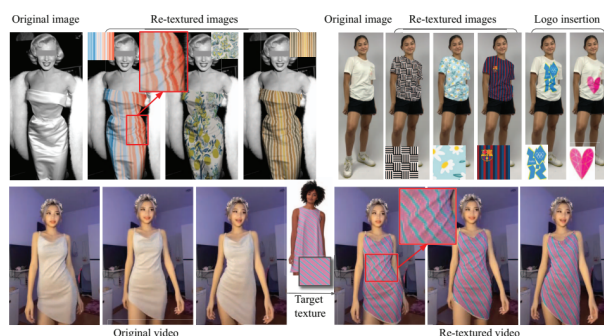


图 1. 基于图像与视频的几何感知纹理编辑。本文提出一种创新方法，可从单张图像（首行）和视频（次行）中预测服装的几何感知纹理（UV）映射图。通过利用三维表面法线，预测的 UV 映射图能保持纹理空间与三维表面之间的等距特性。此外，我们确保视频帧间预测 UV 映射的时序一致性，从而实现物理合理的人物外观编辑。项目网站：yasamin.page/normal-guided-uv。

1. 引言

在浏览在线服装店时，您是否曾想过，如果给定一件形状相似的衣服，某件感兴趣的连衣裙穿在您身上会是什么样子，就像在试衣间里一样？实现这种视觉体验的关键技

术是逼真的重新纹理化 (Re-texturing) ——即根据展示图像或视频中受试者的动作，以几何和时间一致的方式编辑衣服的纹理。在过去的几年里，图像和视频编辑技术取得了重大进展，例如在行驶汽车的视频中插入广告标志或在社交媒体上应用面部妆容。然而，这些针对刚性或半刚性表面设计的编辑方法不适用于相对于底层身体发生复杂次级运动的服装。例如，图 29 中连衣裙的细微褶皱会导致纹理随时间产生复杂的扭曲。在本文中，我们提出了一种新方法，通过考虑服装的细粒度几何形变，来编辑给定图像或视频中服装的外观。

以前的工作主要通过两种方式解决逼真的纹理编辑：(1) **3D 重建和渲染**：这些方法在给定高度精确的 3D 几何形状时可以实现高保真纹理编辑。另一方面，它们的性能取决于 3D 重建的质量。虽然服装的 3D 几何形状可以从成对的人体外观数据（例如具有 3D 网格和渲染的人体建模库）中学习，但由于此类数据的稀缺，它通常无法很好地泛化到从未见过的真实图像和视频。中。(2) **直接纹理映射**：通过估计密集 UV 贴图，这些方法可以绕过 3D 重建的过程。然而，它们通常缺乏几何细节，仅捕捉底层的身体形状，因此不适用于编辑服装。此外，当应用于视频时，编辑的视觉伪影会变得更加显著，因为它们没有意识到服装 3D 几何形状的底层变形。

我们设计的方法兼具这两种方法的优点：在避免 3D 重建的同时保留 UV 映射中的逼真细节。我们的核心见解是，等距性这一基本几何属性可以通过从图像预测出的 3D 表面法线施加到 UV 贴图估算中。我们将 UV 贴图和表面法线之间的几何关系公式化为一组偏微分方程。

我们的方法输入图像或视频、其表面法线预测以及密集光流（针对视频），并输出几何感知的 UV 贴图估算。UV 贴图由多层感知器 (MLP) 建模，该感知器可以根据图像中的像素位置预测 UV 坐标。我们注意到，UV 贴图的定义取决于参考坐标系的选择。为了消除这种歧义，我们使用预定义的代理 UV 贴图（如 DensePose）来约束神经网络。我们使用等距约束作为损失来优化 UV 贴图。此外，对于视频，我们利用每帧图像特征，通过光流关联跨时间的像素 UV 坐标。

我们的贡献可以总结为三个方面：

- 一种新的公式，通过等距约束捕捉 3D 表面法线与 UV 贴图之间的几何关系，从而消除了 3D 重建和真值 UV 贴图的需求；
- 一种神经网络设计，通过关联每帧图像特征来学习预测帧间时间相干的 UV 贴图；
- 与现有的重新纹理化方法相比，具有更强的性能，并在广泛的真实世界图像上取得了引人注目的结果，如表 4 所示。

2. 相关工作

我们的工作位于从图像进行人体 UV 贴图预测和神经 UV 贴图优化这两个领域的交汇处。

2.1. 人体密集 UV 贴图预测

DensePose 的开创性工作学习预测图像中人体的 UV 贴图，这为在不进行 3D 重建的情况下编辑人的外观开辟了新机会。随后的一系列工作带来了一系列人体追踪应用。然而，由于它们采用了针对身体表面的特定表示，在表达如裙子和连衣裙等高度变形的宽松衣物时表现出根本性的局限。为了应对这一挑战，最近的方法利用多任务学习或引入测地线距离来学习 UV 贴图。BodyMap 结合了视觉 Transformer (ViT) 以在连续的身体表面上学习逐像素图像特征，从而处理宽松的衣服、不同的发型和遮挡。TemporalUV 侧重于通过外推初始的 DensePose 估算并利用从输入视频获得的图像特征来处理服装，从而获得与服装边界对齐的 UV。尽管这些方法很有前景，但由于缺乏对底层 3D 几何形状的理解，视觉伪影仍然存在。与以前的方法不同，我们设计的框架使生成的 UV 贴图满足等距性这一基本几何属性，从而实现物理上合理的重新纹理化。

2.2. 视频中的神经 UV 优化

另一类研究转向分层 UV 贴图，通过结合视频分解在一定程度上捕捉几何形状，从而根据观察到的运动优化前景和背景的 UV 坐标。Kasten 等人将视频展开为一组分层 2D 图集，为视频中的每个像素预测其在每个图集中相应的 2D 坐标。Ye 等人提出了全局精灵图 (Sprite)，可以对不同的运动轨迹进行分组，因为集体对象结构在整个时间内具有一致的外观。虽然这些方法保持了时间相干性并维持了一些与手臂或腿部运动相关的粗糙 UV 变形，但它们无法捕捉细微的变形（如衣物中的皱纹）。此外，这些技术无法表示单张图像的 UV 映射，只能应用于视频。与这些方法不同，我们以表面法线为条件的 UV 贴图对微小的几何细节和 3D 表面形变高度敏感，这不仅可以针对视频进行优化，也可以针对单张图像进行优化。

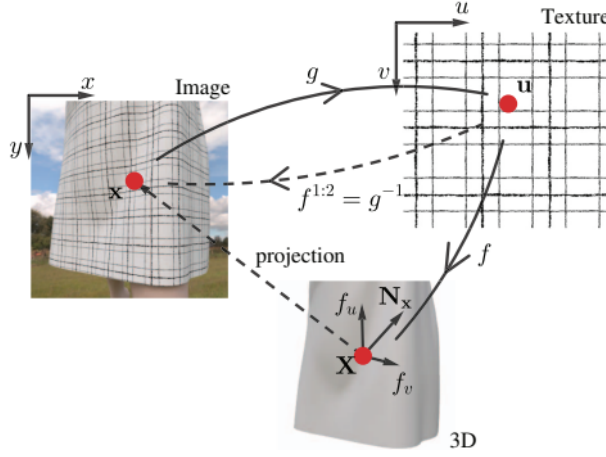


图 2. 映射 $g(x) = u$ 的示意图，它将图像空间中的像素位置映射到服装 UV 空间中的点。

3. 提出的方法

我们的目标是获得连续的纹理映射，从而能够对动态服装的外观进行编辑。我们利用等距映射的几何特性，以偏微分方程的形式约束 UV 贴图。通过优化神经网络来求解这一偏微分方程，从而生成具有几何感知的 UV 贴图。

3.1. 无需三维重建的纹理映射

考虑一个映射 $g(x) = u$ ，它将图像空间中的像素位置 $x = (x, y) \in \mathbb{R}^2$ （属于感兴趣的服装区域）映射到服装 UV 空间中的点 $u = (u, v) \in \mathbb{R}^2$ ，如图 2 所示。我们工作的目标是找到这样一个映射 g ，该映射需考虑通过 x 点预测的表面法向量所度量的局部表面几何信息。我们将 x 点在相机空间中预测的 3D 表面法向量记为 $N_x \in \mathbb{S}^2$ 。

让我们定义一个从 UV 纹理贴图到 3D 表面的等距映射 $f(u) = X$ 。这是不可拉伸布料纹理映射的基本属性：

$$\|f_u\| = \|f_v\| = 1, \quad f_u^T f_v = 0, \quad (12)$$

其中 f_u 和 f_v 分别是函数 f 对 u 和 v 的偏导数。从几何意义上讲， f_u 和 f_v 是三维曲面上的切向量，它们的叉积构成了曲面的法向量：

$$\tilde{N}_x = f_u \times f_v = f_u(g(x)) \times f_v(g(x)) \quad (13)$$

其中 $\tilde{N}_x \in \mathbb{S}^2$ 是位于 X 处对应 x 点的表面法向量。

我们可以通过匹配由公式 (2) 推导出的表面法向量 $\tilde{N}_x$ 与图像预测的表面法向量 N_x

来找到 UV 映射 g :

$$\begin{aligned} \text{最小化}(\theta_g, \theta_f) \quad & \sum_x \|f_u(g(x)) \times f_v(g(x)) - N_x\|^2, \\ \text{s.t.} \quad & \|f_u\| = \|f_v\| = 1, f_u^T f_v = 0, \end{aligned} \quad (14)$$

其中 θ_g 和 θ_f 分别是函数 g 和 f 的参数。求解方程 (3) 的一个关键挑战在于 f 的依赖性要求对表面进行完整的 3D 重建。相反，我们提出了一个新的对偶问题，能够有效解决方程 (3) 而无需找到 f 。

我们利用两个属性从方程 (3) 中消去 f 。首先，我们假设采用正交投影，即 $(x, y) = (X, Y)$ ，其中 $X = [X \ Y \ Z]^T$ 。这使得我们能够通过像素坐标来表示三维导数：

$$g = (f^{1:2})^{-1}, \quad (15)$$

其中 $f^{1:2}$ 表示 f 的前两个元素 $((X, Y))$ 。利用反函数定理，函数 g 的雅可比矩阵 J_g 与 f 的关系为：

$$f^{1:2} \circ g(x) = x \rightarrow J_g = (J_{f^{1:2}})^{-1}, \quad (16)$$

通过联立方程 (1) 和 (5)，在消除 f （三维重建）变量后，可将方程 (2) 改写为以下约束条件：

$$\|g_x\| = \sqrt{1 + \frac{\tilde{n}_x^2}{\tilde{n}_z^2}}, \quad \|g_y\| = \sqrt{1 + \frac{\tilde{n}_y^2}{\tilde{n}_z^2}}, \quad g_x^T g_y = \frac{\tilde{n}_x \tilde{n}_y}{\tilde{n}_z^2}, \quad (17)$$

其中 $\tilde{N}_x = [\tilde{n}_x \ \tilde{n}_y \ \tilde{n}_z]^T$ 。如图 31 所示，这捕捉了曲面与 UV 贴图之间的等距关系。由此产生几何损失函数：

$$\begin{aligned} \mathcal{L}_{\text{geo}}(\theta_g) = \sum_x & \left(\left(\frac{\partial u}{\partial x} \Big|_x \right)^2 + \left(\frac{\partial v}{\partial x} \Big|_x \right)^2 - 1 - \frac{n_x^2}{n_z^2} \right)^2 \\ & + \left(\left(\frac{\partial u}{\partial y} \Big|_x \right)^2 + \left(\frac{\partial v}{\partial y} \Big|_x \right)^2 - 1 - \frac{n_y^2}{n_z^2} \right)^2 \\ & + \left(\frac{\partial u}{\partial x} \Big|_x \frac{\partial u}{\partial y} \Big|_x + \frac{\partial v}{\partial x} \Big|_x \frac{\partial v}{\partial y} \Big|_x - \frac{n_x n_y}{n_z^2} \right)^2, \end{aligned} \quad (18)$$

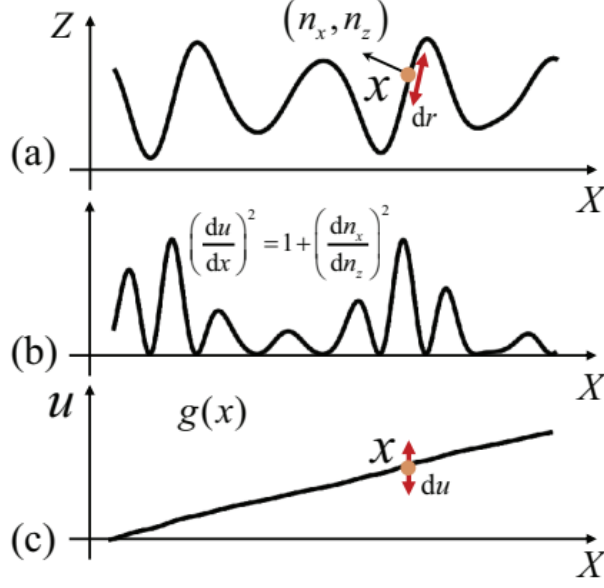


图 3. UV 映射估计的二维简化模型。我们展示了 (a) 曲面与 (c) UV 贴图之间的等距关系。在映射到 $g(x)$ 时， XZ 平面中的曲线长度需要保持不变，即 $|du| = |dr| = \sqrt{dx^2 + dz^2}$ （弧长守恒）。该关系可以利用公式 (6)，通过表面法线和 g 的空间导数的倒数改写为 (b) 偏微分方程形式。

3.2. 纹理映射的自监督学习

纹理映射 g 的定义取决于一个双射函数。我们通过寻找满足 $g \approx g'$ 的函数 g 来消除这种歧义，其中 g' 是预定义的人体代理映射：

$$\mathcal{L}_{\text{prox}}(\theta_g) = \sum_x \|g'(x) - g(x)\|^2 \quad (19)$$

在实践中，我们采用扩展版 DensePose 作为预定义代理映射。此外，我们通过添加以下损失函数来确保可见 3D 表面的物理合理性，即纹理贴图应使表面法线指向 +Z 方向：

$$\mathcal{L}_z(\theta_g) = \sum_x \max(0, -\det(J_g|_x)) \quad (20)$$

对于视频，我们利用跨帧的光流技术来确保纹理贴图的时间一致性：

$$\mathcal{L}_{\text{tmp}}(\theta_g) = \sum_{i,j} \sum_{x_i} \|g(x_i, f_{x_i}) - g(x_j, f_{x_j})\|^2 \quad (21)$$

综上所述，我们通过优化以下损失函数来学习纹理贴图：

$$\mathcal{L}(\theta_g) = \mathcal{L}_{\text{geo}} + \lambda_{\text{prox}} \mathcal{L}_{\text{prox}} + \lambda_z \mathcal{L}_z + \lambda_{\text{tmp}} \mathcal{L}_{\text{tmp}} \quad (22)$$

其中，当使用单张图像时， $\lambda_{\text{tmp}} = 0$ 。

3.3. 实现细节

我们采用一个 12 层的多层感知器对 g 进行建模（如图 32 所示），并在每层后使用 ReLU 作为激活函数。像素坐标经过基于傅里叶的位置编码 $\gamma: \mathbb{R}^2 \rightarrow \mathbb{R}^{128}$ 作为输入。针对视频处理（图 33），我们使用 ResNet 提取每帧 256 维的图像特征 f_x 。我们使用 Adam 优化器，批次大小为 2048，学习率为 10^{-4} 。本方法处理 82 帧视频耗时 20 分钟，显著快于表 4 中对比的现有基准方法。

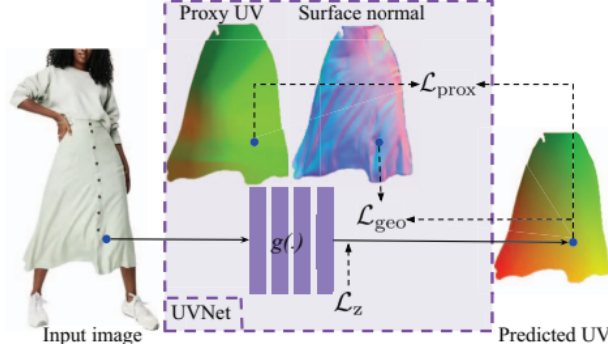


图 4. 单图像框架。针对服装上的每个像素位置 x ，我们通过多层感知器预测 UV 坐标 u 。通过将预测的 3D 表面法线与 UV 映射进行匹配，我们强制实现等距映射。 $\mathcal{L}_z$ 确保预测的表面法线指向相机。为消除 UV 映射参考框架的歧义，我们采用预定义的代理映射（如 DensePose 外推法）并应用 $\mathcal{L}_{prox}$ 损失函数。

4. 评估

我们在真实图像以及带有真实 UV 映射的合成数据上，从定量和定性两个维度评估了我们的方法。同时，我们还与人体 UV 映射估计方法和视频 UV 映射优化领域的最新先进技术进行了对比。

4.1. 评估数据集和指标

评估数据集：我们采用以下数据集对我们的方法进行评估：(1) 来自 Santesteban 等人的 5 段合成视频序列，包含模拟连衣裙和 T 恤服装，带有随机纹理图案，总帧数超过 700 帧；(2) 来自 Fashion Video 数据集的 10 段真实视频；(3) TikTok 数据集以及各类 YouTube 视频和网络野生图像。

评估指标：我们采用五项指标评估本方法：(1) **UV 误差 (UV error)**：针对合成数据，我们在纹理空间中报告绝对 UV 误差。通过普氏分析 (Procrustes analysis) 对齐生成的纹理贴图以消除参考系歧义。以公制刻度报告均方误差（表 4）。(2) **平均精度百分比 (AP)**：计算所有像素的平均精度；若预测 UV 值与真实值的误差低于阈值，则判定预测

正确。针对 1 至 15 厘米范围内的多个阈值可视化的 AP 指标如图 18 所示。(3) 光度误差 (**Photometric error**): 我们使用 UV 贴图估计将输入视频的首帧扭曲至后续帧, 并报告真实 RGB 图像与扭曲后 RGB 图像之间的误差 (表 4)。(4) 几何误差 (**Geometric error**): 我们报告 $\mathcal{L}_{\text{geo}}$ 指标, 以衡量预测的 UV 贴图遵循表面法线估计中捕获的几何信息的能力。(5) 时序误差 (**Temporal error**): 我们通过计算 $\mathcal{L}_{\text{tmp}}$ 误差来评估不同方法在保持时序连贯性方面的表现。

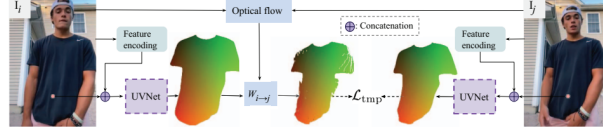


图 5. 视频框架。我们利用光流在预测的 UV 映射中应用时间一致性。借助光流 $W_{i \rightarrow j}$, 我们通过时间损失 $\mathcal{L}_{\text{tmp}}$ 将第 i 帧的 UV 预测结果与第 j 帧进行匹配。



图 6. 近邻损失 $\mathcal{L}_{\text{prox}}$ 对 UV 预测与重新纹理化的影响。

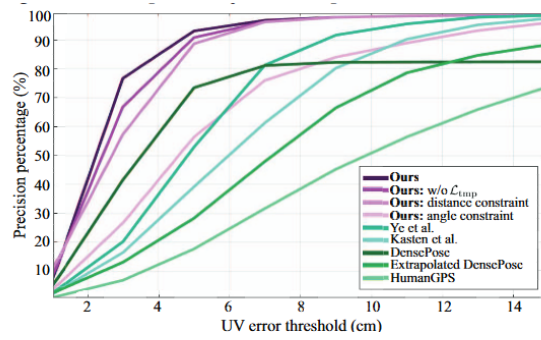


图 7. 平均精度。我们在真实服装序列上, 计算了我们的方法、基准方法以及消融实验中, 紫外误差高于阈值 (1-15 厘米) 的像素的平均值。

表 1. 定量结果。分别为紫外误差（UV.error，单位：厘米）、光度误差（photo. error，即 RGB 差值）、几何损失（geo. error, $\mathcal{L}_{geo}$ ），以及时间损失（tmp. error, $\mathcal{L}_{tmp}$ ）（图像空间像素距离），以（均值 $\pm$ 标准差）呈现。

Method	GT dress sequences [45]		GT T-shirt sequences [45]		Real Fashion sequences [59]	
	UV. error (cm)	photo. error	UV. error (cm)	photo. error	geo. error ($\mathcal{L}_{geo}$)	tmp. error
DensePose [18]	17.27 $\pm$ 3.76	52.43 $\pm$ 10.20	7.48 $\pm$ 0.52	34.19 $\pm$ 7.01	1.26 $\pm$ 0.47	8.12 $\pm$ 2.70
Extrapolated DensePose [18, 50]	8.61 $\pm$ 0.76	18.29 $\pm$ 3.24	5.34 $\pm$ 0.63	18.86 $\pm$ 4.55	0.52 $\pm$ 0.06	4.62 $\pm$ 0.73
HumanGPS [47]	11.97 $\pm$ 2.06	97.41 $\pm$ 30.87	7.53 $\pm$ 0.43	108.10 $\pm$ 34.89	1.27 $\pm$ 0.69	51.40 $\pm$ 41.66
Kasten et al. [25]	7.06 $\pm$ 0.60	13.07 $\pm$ 2.79	6.64 $\pm$ 0.63	14.54 $\pm$ 4.26	0.56 $\pm$ 0.10	2.30 $\pm$ 0.70
Ye et al. [57]	5.56 $\pm$ 0.29	33.57 $\pm$ 10.27	5.75 $\pm$ 0.20	19.22 $\pm$ 5.14	0.71 $\pm$ 0.03	1.65 $\pm$ 0.15
Ours	3.16$\pm$0.28	7.54$\pm$2.04	3.58$\pm$0.27	11.28$\pm$2.01	0.07$\pm$0.03	1.50$\pm$0.23

表 2. 服装序列的消融研究。结果包含紫外误差（UV.error，单位：厘米）与光度误差（photo. error，即 RGB 差值），以“均值 $\pm$ 标准差”呈现。

Method	UV. error (cm)	photo. error (RGB)
Ours	3.16$\pm$0.28	7.54$\pm$2.04
Distance constraint	3.44 $\pm$ 0.25	8.29 $\pm$ 1.99
Angle constraint	5.96 $\pm$ 0.33	7.68 $\pm$ 1.37
W/o $\mathcal{L}_{tmp}$	3.22 $\pm$ 0.24	14.67 $\pm$ 4.82
W/o $\mathcal{L}_{prox}$	3.24 $\pm$ 0.35	8.46 $\pm$ 2.19

4.2. 消融研究

我们通过消融实验分析了距离和角度约束的影响，结果汇总于表 5 和图 18。我们的最终方法在 UV 误差和光度误差方面均取得最佳性能。正如预期的那样，去除时间一致性约束（ $\mathcal{L}_{tmp}$ ）会导致光度误差表现显著较差，因为未强制执行帧间一致性。 $\mathcal{L}_{prox}$ 的作用是消除等效 UV 映射的歧义性；正如图 34 所示，不使用 $\mathcal{L}_{prox}$ 的重纹理化可能会导致不同对象出现任意的朝向。

4.3. 定性结果展示和用户研究

表 3. 用户研究结果。

Method	User score
DensePose	4.09 ± 1.94
Proxy DensePose	3.23 ± 1.86
HumanGPS	2.28 ± 2.12
Kasten et al.	5.52 ± 1.88
Ye et al.	7.66 ± 1.27
Ours	9.57 ± 0.81

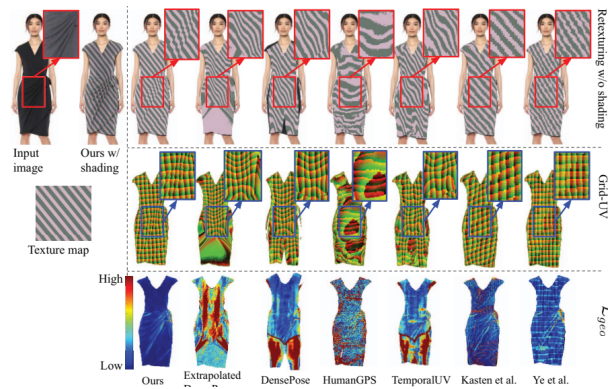


图 8. 定性对比。我们将我们的方法与初始 UV（外推的 DensePose）、DensePose、HumanGPS、Kastenn 等人的方法以及 Ye 等人的方法，在图像重新纹理化与 UV 映射网格可视化方面进行了对比。



图 9. 局限性。由于缺少三维重建的表面，我们的方法无法处理会导致纹理不连续的褶皱。我们的纹理映射在褶皱区域周围是连续的，这在物理层面是不合理的。

为了直观呈现我们的效果，我们展示了重着色示例和网格 UV 示意图，以体现各方法在保留高频细节方面的表现。图 19 展示了在时尚视频序列上的结果，而图 22 展示了本方法在视频和图像上的表现效果。我们的方法不仅能捕捉细粒度表面细节，还能保持跨时间序列的时序一致性。

我们还开展了一项用户研究：邀请 21 名参与者对我们的方法与基准方法进行真实感评分（1 分代表极不真实，10 分代表非常真实）。如表 6 所示，我们的方法获得了用户给出的最高评分（ 9.57 ± 0.81 ）。

5. 结论

本文提出了一种创新方法，通过保留图像和视频中的几何细节来预测高质量 UV 贴图。我们利用三维表面法线中编码的等距几何特性，以偏微分方程形式优化 UV 贴图。通过整合光流技术，我们将该方法推广至视频处理，实现了时序连贯的视频编辑效果。



图 10. 局限性。带纹理服装上的局限性。



图 11. 图像与视频上的结果。在给定图像和视频的情况下，我们的方法能够生成出色的纹理编辑效果，同时保留几何细节。

局限性 正如 3.1 节所述，关于投影的一个假设是，三维表面几何与图像之间存在一一对应关系。然而，当存在褶皱导致部分三维表面区域在图像中不可见时，这一假设就不成立了。图 20 展示了这种局限性：裙子出现褶皱时，纹理必须是不连续的，而我们的方法因缺乏三维重建几何结构而生成连续纹理渲染，这在物理上是不准确的。

另一个局限性在于具有高对比度纹理的服装。着色操作往往容易受到颜色对比度偏差的影响，导致纹理区域附近的外观显得不真实。图 21 展示了尽管 UV 预测合理，但在复杂图案附近的最终生成效果仍不理想。改进着色操作超出了本研究的范围，我们将以此作为未来的工作。

Color-Mood-Aware Clothing Re-texturing

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Abstract: In this paper, we present a novel color-mood-aware technique to re-texture clothing in a photograph. An efficient classification algorithm is developed to classify clothing textures using color mood scheme. To re-texture the clothing, our approach first computes the gradient maps for the cloth region to be replaced and then calculates the texture distortion coordinates on the projected cloth region according to the gradient maps. After the user selects a target clothing texture from the classified clothing texture database, the lighting and shading effects on the original photograph is transferred using the HSV color space. Experimental results show that the proposed approach successfully re-textures the clothes in photographs while preserving the geometry and lighting features.



Clothing re-texturing using color mood scheme:(a) the input photograph(the re-texturing cloth region(with red boundary));(b)-(d) the re-texturing results. Note that the input clothing textures are the insets in(b)-(d)(left bottom).

1. Introduction

Clothing re-texturing is a process to replace the existing clothing texture in the concerned region of a photograph by new materials while preserving the original shading and lighting effects for e-entertainment multimedia applications. Although there exists a rich body of work on image texturing and re-texturing, the clothing re-texturing receives relatively little attention. Given a single photograph capturing a person wearing a piece of cloth (e.g. a skirt or a t-shirt), we want to allow users to simulate the person's looking and mood when dressed in clothes of different textures through some texture replacing techniques. For this purpose, we develop a new texture classification algorithm based on color mood scheme.

We also developed an improved texture replacement algorithm for replacing the original clothing texture with a correctly deformed and shaded texture of personalized mood. Our approach requires no 3D models, and can generate the highly realistic final results. It has been a long history for the research on relationships between colors and their evoked moods. Whelan et al. developed a color mood scheme, which defines the 24 mood categories and their primary

corresponding colors. However, the current approaches do not address the problem on how to calculate which color mood it belongs to when given a single clothing texture. In our paper, we developed such a novel classification method for clothing textures using the histogram-based color mood scheme. Figure 29 shows the clothing re-texturing results with our approach using the Trendy mood, Traditional mood, and Earthy mood, respectively.

2. Clothing texture classification

2.1. Color mood scheme

In order to create an efficient mood table to guide the clothing texture classification, we adopt the primary color-based histogram to denote the color mood. Figure 30 shows the 24 mood categories and their primary corresponding colors. The employed color mood scheme encompasses a broad range of color moods and is thus suitable for texture classification.

For instance, the Tropical mood often expresses traditional themes such as richness and stability, which is frequently seen in the decor of banks and legal offices, suggesting permanence and value. The Vital mood usually expresses a feeling of vigor and warmth, it is youthful and playful and is often seen in advertisements displaying energetic lifestyles and personalities. The Dependable mood is interpreted as dependable and reliable, communicating firmness and strength.

2.2. Color mood guided classification

For the purpose of easy clothing re-texturing, we develop an efficient clothing texture classification algorithm based on the color mood. In order to classify the input textures into one of the 24 color moods, we employ the mood histogram to determine which mood it belongs to. Our color mood histogram is defined as:

$$h(c_k) = n_k, \quad k \in [0, 23] \quad (23)$$

Here c_k is the k^{th} color mood in the mood interval [c_0 = Powerful mood, c_1 = Rich mood, $\dots$, c_{23} = Professional mood] and n_k is the number of pixels in the clothing texture whose color mood is c_k . In our application, the color mood histogram is calculated for each sample clothing texture in the database. Each pixel of the clothing texture is compared with all 24 moods to find the closest mood, then the pixel is added to that color mood's bin c_k in the color histogram $h(c_k)$. Here, the Euclidean distance metric is employed for measuring the closest mood in our scheme.

In order to classify a clothing texture into some corresponding color mood, we determine its category by locating the mood with the highest frequency in its histogram, as illustrated in Figure 31. A similar color mood histogram scheme is used for color transfer between images. Figure 32 shows some results classified by the color mood histogram approach in our clothing texture database.



Figure 2. The 24 color moods and their corresponding primary colors.

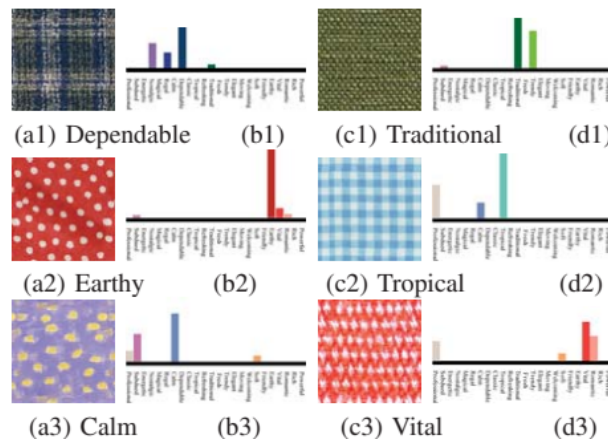


Figure 3. Classified color moods and their corresponding mood-histograms:(a1)-(a3),(c1)-(c3), the input textures;(b1)-(b3),(d1)-(d3), the mood-histograms.



Figure 4. Classified clothing texture database using color mood scheme.

3. Clothing re-texturing approach

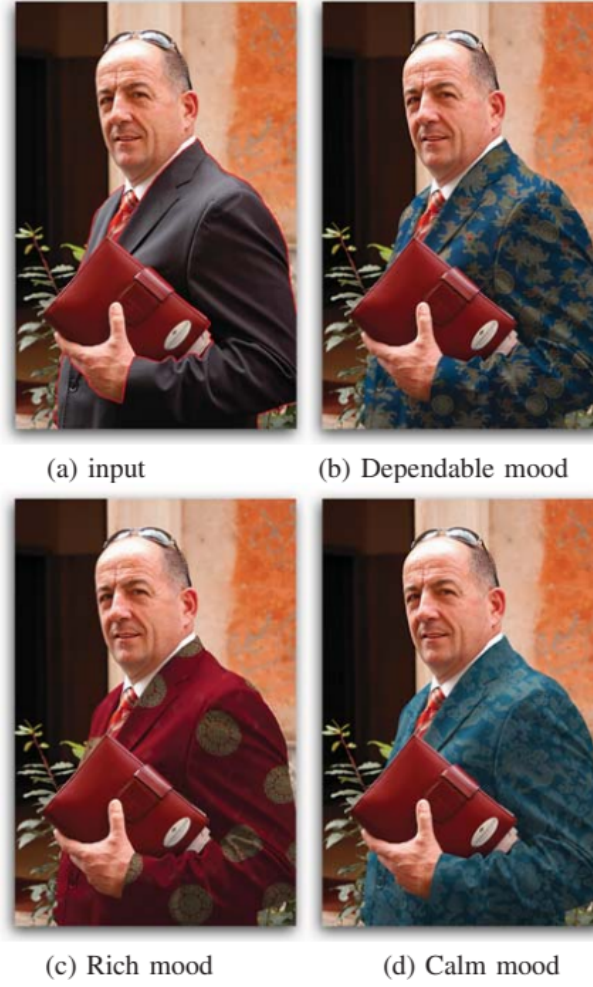


Figure 5. Results by our algorithm:(a) the input photograph;(b)-(d) the re-texturing results.

The color mood guided clothing re-texturing approach consists of five steps, which are outlined as follows:

1. Calculate the gradient map: $\nabla I_G(x, y) = \left(\frac{I_L(x+1, y) - I_L(x-1, y)}{2}, \frac{I_L(x, y+1) - I_L(x, y-1)}{2} \right)$. Here, I_L is the luminance image of the input photograph with the replaced region Ω .
2. Smooth the gradient map $\nabla I_G(x, y)$ by applying the bilateral filter, and get the filtered gradient map $\nabla I_G^S(x, y)$.
3. Compute the texture indices $[t_x, t_y]$ according to the smoothed gradient map $\nabla I_G^S(x, y)$.
4. Derive the new color $I'(x, y)$ from its original color and the input color mood clothing texture $T(t_x, t_y)$.

5. Transfer the lighting and shading effect from the original photograph in the HSV color space.

Our approach focuses on computing a locally-consistent gradient maps, and then we use these gradients to derive the texture distortion for the re-texturing region. When an input photograph I is given, we first compute the gradient field $\nabla I_G(x, y)$ in terms of neighboring luminance values of its luminance image I_L :

$$\nabla I_G(x, y) = \left(\frac{I_L(x+1, y) - I_L(x-1, y)}{2}, \frac{I_L(x, y+1) - I_L(x, y-1)}{2} \right) \quad (24)$$

We use the gradient bilateral filter (GBF) to smooth the gradient map $\nabla I_G(x, y)$. Here, the gradient bilateral filter at pixel p is defined by:

$$K = \sum_{q \in \nabla I_G} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}(\|\nabla I_G^p - \nabla I_G^q\|) \quad (25)$$

$$\nabla I_G^S(x, y) = GBF(\nabla I_G(x, y)) = \frac{1}{K} \sum_{q \in \nabla I_G} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}(\|\nabla I_G^p - \nabla I_G^q\|) \nabla I_G^q$$

where σ_s controls the spatial neighborhood, σ_r controls the influence of the intensity difference, $G_{\sigma_r}(\cdot)$ and $G_{\sigma_s}(\cdot)$ are the kernels in the form of the Gaussian function.

3.1. Mood-aware clothing re-texturing

In general, the recovered gradient field is sufficient to be used to estimate a warping of a color mood-aware clothing material texture T . Our approach employs the lazy snapping technique to segment the cloth region. Unlike other methods, our algorithm does not need to generate a proper triangle mesh for calculating the texture distortion coordinates. For extra flexibility, the smoothed gradient map is introduced to calculate the texture indices $[t_x, t_y]$ by:

$$\nabla t_x = (1 + k_1) * \nabla I_G^S, \quad \nabla t_y = (1 + k_2) * \nabla I_G^S \quad (26)$$

$$\Delta t_x(x, y) = \text{div} \nabla t_x(x, y), \quad \Delta t_y(x, y) = \text{div} \nabla t_y(x, y) \quad (27)$$

where k_1 and k_2 are the scale parameters, Δ and div represent the Laplacian and divergence operator, respectively. We adopt the Poisson equation to solve them and it runs very fast.

The new color value $I'(x, y)$ is derived from its original color and the color of the mood-aware clothing texture $T(t_x, t_y)$ with the following matting equation:

$$I'(x, y) = (1 - f) * T(t_x(x, y), t_y(x, y)) + f * I(x, y) \quad (28)$$

where f is a scalar parameter that linearly interpolates the original cloth color and the replaced clothing texture color. Due to the lack of simulating the lighting and shading effects, the result looks flattening if only the Poisson equation is used. To make the realistic appearance, the final relighting result $I''(x, y)$ is obtained by copying the H and S components of the replaced texture and blending the V component:

$$I''_H(x, y) = I'_H(x, y), \quad I''_S(x, y) = I'_S(x, y) \quad (29)$$

$$I''_V(x, y) = (1 - k_3) * I'_V(x, y) + k_3 * I_V(x, y) \quad (30)$$

where k_3 defines the balancing weight between the replaced clothing texture and the brightness of the original photograph. Results of this approach are shown in Figure 33 and Figure 34.



Figure 6. Experimental results by our approach:(a)the input photograph(the re-clothing region(with red boundary));(b)-(e) the re-texturing results using the Subdued mood, Calm mood, Trendy mood, Elegant mood, and Calm mood, respectively.

4. Experimental results

We have applied our proposed algorithm to a variety of photographs and the experimental results demonstrated that our clothing re-texturing technique can generate visually pleasing results. Figure 34 shows some experimental results of our clothing re-texturing algorithm. The example shows, in order from left to right, the input photograph, the replacement result by Dependable mood (Figure 34(b)), Rich mood (Figure 34(c)), Calm mood (Figure 34(d)), and Regal mood (Figure 34(e)), respectively.

Figure 29 shows more re-texturing results using the textile material textures. As shown in Figure 29 and Figure 33, the cloth ripples and wrinkles are faithfully transferred in the final clothing replacement results. Additionally, our approach produces similar shadow and illumination effects on the cloth region as that of the input photograph. Moreover, we can combine our method with other augmented applications, such as the virtual fashion show and the augmented reality system on cloth with realistic illumination.

5. Conclusion and discussions

We have presented a novel color-mood-aware clothing re-texturing algorithm for multimedia applications. Our approach computes the gradient maps for the region to be replaced, calculates the texture distortion coordinates on the projected region of the underlying surface according to the gradient maps. The lighting and shading effect on the original photograph is transferred using the HSV color space.

Given a set of clothing material, we calculate the color mood histogram for each texture, and build a database of clothing textures classified by color mood. Thus, the user can easily select the target clothing texture of preferred mood from the classified clothing texture database. Our system is better suited for the virtual cloth show and simulation application in the entertainment industry, where the user can easily simulate the visual effect of different cloth.

6. Acknowledgements

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译文 2

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色彩情绪感知的衣物重纹理技术

摘要: 本文提出一种新颖的色彩情绪感知技术, 用于对照片中的服装进行纹理替换。我们开发了一种高效的分类算法, 采用色彩情绪方案对服装纹理进行分类。在重新纹理化过程中, 该方法首先计算待替换布料区域的梯度图, 随后根据梯度图在投影布料区域上计算纹理形变坐标。当用户从分类好的服装纹理库中选择目标纹理后, 通过 HSV 色彩空间迁移原始照片中的光照与阴影效果。实验结果表明, 所提出的方法在保持几何结构和光照特征的同时, 成功实现了照片中服装的纹理重构。



图 1. 使用颜色情绪方案进行服装重纹理: (a) 输入照片 (待重纹理的服装区域以红色边界标出); (b) - (d) 重纹理结果。请注意, 输入的服装纹理是 (b) - (d) 子图中左下角的插图。

1. 引言

时尚图像编辑是指通过将所需的时尚物品（如不同的服装）以现实且视觉上令人信服的方式融入到给定图像中的任务。成功应用此类算法可以让用户在家里舒适地试穿选定的服装。此功能有潜力促进在线服装销售，减少零售商成本，并通过最小化退货，降低时尚行业的环境影响。因此，近年来，越来越多的研究努力集中在时尚图像操控（或虚拟试穿 VTON）技术上，这些技术能在多年的发展中提供令人信服的逼真编辑效果。

服装重纹理是一种将照片中特定区域的现有服装纹理替换为新材质的过程，同时需保留原始的阴影和光照效果，旨在用于电子娱乐多媒体应用。尽管在图像纹理化和重纹理化方面已有大量研究工作，但服装重纹理所受到的关注相对较少。给定一张捕捉到穿着某件衣物（例如，裙子或 T 恤）的人像照片，我们希望通过一些纹理替换技术，让用户能够模拟人物穿着不同纹理服装时的外观和情绪。为此，我们开发了一种基于颜色情绪方案的新纹理分类算法。

我们还改进了一种纹理替换算法，用于将原始服装纹理替换为经过正确变形和着色、且具有个性化情绪的纹理。我们的方法无需 3D 模型，并能生成高度逼真的最终结果。关于颜色与其所引发情绪之间关系的研究已有很长历史。Whelan 等人开发了一种颜色情绪方案，该方案定义了 24 种情绪类别及其主要对应颜色。然而，现有方法并未解决在给定单一服装纹理时，如何计算其所属颜色情绪的问题。在我们的论文中，我们开发了这样一种新颖的、基于直方图的颜色情绪方案对服装纹理进行分类的方法。图 29 展示了使用我们的方法，分别采用时尚情绪（图 1(b)）、传统情绪（图 1(c)）和质朴情绪（图 1(d)）进行服装重纹理的结果。

2. 服装纹理分类

2.1. 颜色情绪方案

为了创建一个高效的情绪表来指导服装纹理分类，我们采用基于主色相的直方图来表示颜色情绪。图 30 展示了 24 种情绪类别及其对应的主色相。所采用的色彩情绪方案涵盖了广泛的颜色情绪范围，因此适用于纹理分类。

例如，热带情绪（图 3c2）通常表达传统主题，如丰富和稳定，常见于银行和法律办公室的装饰中，暗示着永久性和价值。活力情绪（图 3c3）通常表达一种活力和温暖的感觉，它年轻且有趣，常见于展示充满活力的生活方式和个性的广告中。可靠情绪（图 3a1）被解读为可靠和可信赖，传达了坚定和力量。我们建议读者参阅 Whelan 的著作以获取更多关于颜色情绪及其情感的信息。

2.2. 颜色情绪指导的分类

为了实现便捷的衣物重纹理处理，我们基于色彩情绪开发了一种高效的服装纹理分类算法。为了将输入纹理划分至 24 种色彩情绪类别之一，我们采用情绪直方图判定其所属情绪类别。我们的色彩情绪直方表定义为：

$$h(c_k) = n_k, \quad k \in [0, 23] \quad (31)$$

其中 c_k 代表情绪区间内的第 k 种色彩情绪 [c_0 = 强烈情绪, c_1 = 浓郁情绪, ..., c_{23} = 专业情绪], 而 n_k 表示服装纹理中色彩情绪为 c_k 的像素数量。在我们的应用中，会为数据库中的每个样本服装纹理计算色彩情绪直方图。将服装纹理的每个像素与全部 24 种情绪进行比对以找出最接近的情绪，随后将该像素计入色彩直方图 $h(c_k)$ 中对应色彩情绪 c_k 的区间。本方案采用欧几里得距离度量来计算最接近的情绪。

为了将服装纹理分类到相应的颜色情绪，我们通过定位其直方图中频率最高的情绪来确定其类别，如图 31 所示。类似的颜色情绪直方图方案被用于图像间的颜色迁移。图 32 展示了我们的服装纹理数据库中通过颜色情绪直方图方法分类的一些结果。



图 2. 24 种颜色情绪及其对应的主色。

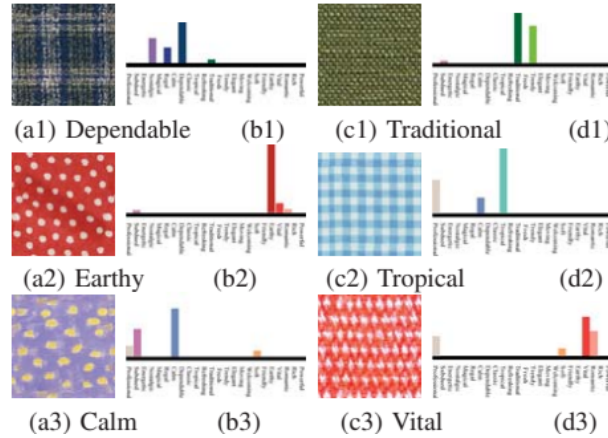


图 3. 分类后的颜色情绪及其对应的情绪直方图：(a1) - (a3) 为输入纹理；(b1) - (b3) 为情绪直方图。



图 4. 使用颜色情绪方案分类的服装纹理数据库。

3. 服装重纹理化方法

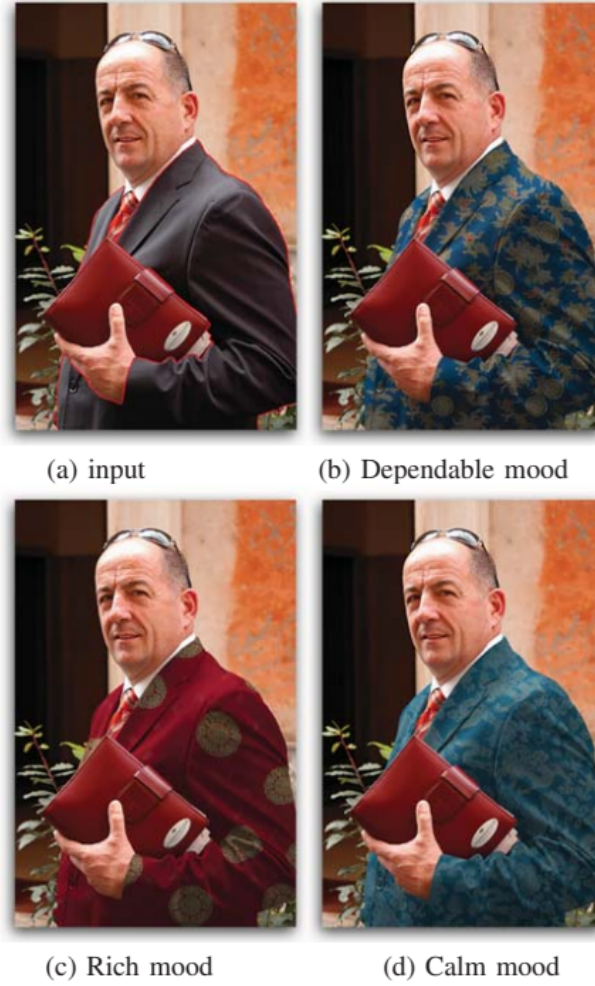


图 5. 我们的算法结果：(a) 输入照片；(b) - (d) 重纹理结果。

基于色彩情绪引导的衣物重纹理方法包含以下五个步骤：

1. 计算梯度图： $\nabla I_G(x, y) = \left(\frac{I_L(x+1, y) - I_L(x-1, y)}{2}, \frac{I_L(x, y+1) - I_L(x, y-1)}{2} \right)$ 。此处， I_L 为输入照

片中替换区域 Ω 的亮度图像。

2. 对梯度图 $\nabla I_G(x, y)$ 应用双边滤波进行平滑处理，得到滤波后的梯度图 $\nabla I_G^S(x, y)$ 。
3. 根据平滑后的梯度图 $\nabla I_G^S(x, y)$ 计算纹理索引 $[t_x, t_y]$ 。
4. 根据原始颜色与输入色彩情绪服装纹理 $T(t_x, t_y)$ 推导出新颜色 $I'(x, y)$ 。
5. 在 HSV 色彩空间中迁移原始照片中的光照与阴影效果。

我们的方法聚焦于计算局部一致的梯度图，随后利用这些梯度推导出重新纹理化区域的纹理形变。当输入一张照片 I 时，我们首先根据其亮度图像 I_L 的相邻亮度值计算梯度场 $\nabla I_G(x, y)$ ：

$$\nabla I_G(x, y) = \left(\frac{I_L(x+1, y) - I_L(x-1, y)}{2}, \frac{I_L(x, y+1) - I_L(x, y-1)}{2} \right) \quad (32)$$

我们采用梯度双边滤波器 (GBF) 对梯度图 $\nabla I_G(x, y)$ 进行平滑处理。此处，像素点 p 处的梯度双边滤波器定义为：

$$K = \sum_{q \in \nabla I_G} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}(\|\nabla I_G^p - \nabla I_G^q\|) \quad (33)$$

$$\nabla I_G^S(x, y) = GBF(\nabla I_G(x, y)) = \frac{1}{K} \sum_{q \in \nabla I_G} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}(\|\nabla I_G^p - \nabla I_G^q\|) \nabla I_G^q$$

其中 σ_s 控制空间邻域范围， σ_r 控制强度差异的影响程度， $G_{\sigma_r}(\cdot)$ 和 $G_{\sigma_s}(\cdot)$ 是采用高斯函数形式的核函数。

3.1. 情绪感知的服装重纹理

总体上，恢复的梯度场足以用于估算色彩情绪感知服装材质纹理 T 的形变。本方法采用懒人分割技术对衣物区域进行分割。与以往方法不同，我们的算法无需生成特定三角网格来计算纹理畸变坐标。为增强灵活性，引入平滑后的梯度图通过计算纹理索引 $[t_x, t_y]$ 来实现：

$$\nabla t_x = (1 + k_1) * \nabla I_G^S, \quad \nabla t_y = (1 + k_2) * \nabla I_G^S \quad (34)$$

$$\Delta t_x(x, y) = \text{div} \nabla t_x(x, y), \quad \Delta t_y(x, y) = \text{div} \nabla t_y(x, y) \quad (35)$$

其中 k_1 和 k_2 为尺度参数， Δ 和 div 分别代表拉普拉斯算子与散度算子。我们采用泊松方程进行求解，其运行速度极快。

新颜色值 $I'(x, y)$ 由其原始颜色与情绪感知服装纹理 $T(t_x, t_y)$ 的颜色通过以下抠图方程推导得出：

$$I'(x, y) = (1 - f) * T(t_x(x, y), t_y(x, y)) + f * I(x, y) \quad (36)$$

其中 f 是一个标量参数，用于线性插值原始衣物颜色与替换后的服装纹理颜色。由于缺乏对布料区域光照与阴影效果的模拟，若仅采用上述泊松方程推导出的纹理扭曲坐标，替换后的结果会显得平面化。为使替换后的服装区域呈现真实外观，必须妥善传递光照与阴影效果。最终的重光照结果 $I''(x, y)$ 可通过以下方式获得：将替换服装纹理的 H 、 S 分量复制至目标照片的布料区域，并对替换纹理与原始照片的 V 分量进行加权融合：

$$I''_H(x, y) = I'_H(x, y), \quad I''_S(x, y) = I'_S(x, y) \quad (37)$$

$$I''_V(x, y) = (1 - k_3) * I'_V(x, y) + k_3 * I_V(x, y) \quad (38)$$

其中 k_3 定义了替换衣物纹理与原始照片上布料区域亮度之间的平衡权重。该方法的结果如图 33 和图 34 所示。



图 6. 我们方法的实验结果：(a) 输入照片（待换装的区域以红色边界标出）；(b) - (e) 分别为使用内敛情绪、平静情绪、时尚情绪、优雅情绪和平静情绪进行重纹理的结果。

4. 实验结果

我们将提出的算法应用于多种照片，实验结果表明，我们的服装重纹理技术能够生成视觉上令人愉悦的结果。图 34 展示了我们服装重纹理算法的一些实验结果。该示例从左到右依次显示为：输入照片、使用可靠情绪（图 6(b)）、丰富情绪（图 6(c)）、平静情绪（图 6(d)）和皇家情绪（图 6(e)）的替换结果。

图 29 展示了使用纺织材料纹理的更多重纹理结果。如图 29 和图 33 所示，布料涟漪和皱纹被忠实地转移到了最终的服装替换结果中。此外，我们的方法在布料区域上产

生的阴影和光照效果与输入照片中的效果相似。而且，我们可以将我们的方法与其他增强应用相结合，例如具有真实光照的虚拟时装秀和增强现实系统。

5. 结论与讨论

我们提出了一种新颖的、用于多媒体应用的色彩情绪感知服装重纹理算法。我们的方法计算待替换区域的梯度图，根据梯度图计算底层表面投影区域上的纹理形变坐标。原始照片上的光照和阴影效果在 HSV 颜色空间中进行转移。

给定一组服装材料，我们计算每种纹理的颜色情绪直方图，并建立一个按颜色情绪分类的服装纹理数据库。因此，用户可以轻松地分类好的服装纹理数据库中选择偏好情绪的目标服装纹理。我们的系统更适用于娱乐产业中的虚拟服装展示和模拟应用，用户可以轻松模拟不同服装的视觉效果。

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